

Say, Echo, Do

Strategic speech, media echo, and the economics of false alarms

[Author Name]¹

¹[Affiliation, City, Country] • Correspondence: [email]

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Abstract

Financial markets hear three voices: what institutions *say*, what the media *echoes*, and what capital *does*. This paper develops a theory of the disagreement between them and of how to trade it.

The central result explains *false alarms*. An institution that trades while addressing a partly credulous crowd finds it optimal to talk an asset down while buying it, and this happens in an explicit region of three parameters: how much the crowd believes, how much trading moves prices, and how costly misreporting is. A distribution-free identity then shows that the observable covariance between words and deeds decides whether institutional statements should be followed or faded. Media virality determines which kind of manipulation arises, and when trust is learned over time it can cycle and even become chaotic.

On the measurement side, we give exact tools to separate new information from repetition using both timing and meaning, to align text embeddings with market outcomes, and to detect whether money moved before the narrative. Every closed form is confirmed numerically, and a tested open-source implementation accompanies the paper.

Keywords: investor sentiment • strategic communication • market manipulation • Hawkes processes • contrastive learning • path signatures • conformal prediction

JEL: G12, G14, G41, C45, C58 **MSC 2020:** 91G15, 91B44, 60G55, 62M20

Highlights

- Talking an asset down while buying it is *optimal*, not merely possible, when $\varphi^2 < 2\lambda k < \varphi$.
- Whether to trust institutional words is decided by one observable number: the sign of $\text{Cov}(\text{Say}, \text{Do})$.
- Echo breeds false alarms in deep markets and exaggeration in shallow ones.
- Adaptive trust converges only if $\lambda k > 1$; otherwise it cycles and can become chaotic.
- News and echo can be separated exactly by combining timing with meaning.

1 Introduction

A respected bank publishes a gloomy call on a stock. Within hours the headlines multiply and social media repeats them. The price falls, but by less than the noise suggests, and a few weeks later it trades higher than before. Looking back at the disclosures, someone was buying all along.

Practitioners call such episodes *false alarms*, and many suspect that large institutions sometimes create them in order to buy cheaply. Proving intent is neither possible nor necessary. What *is* possible is to measure, as events unfold, the disagreement between three observable voices (Table 1).

Table 1 | The three voices of a market.

Voice	What it is	Where it is observed
Say	what institutions publicly claim	ratings, target prices, research notes, management tone
Echo	what the media repeats and amplifies	article volume, syndication, social posts, emotional intensity
Do	what capital actually does	insider trades, holdings, futures positioning, options flow, price

Figure 1a sketches the pattern we want to understand. Positioning builds first. Then the alarm arrives, the price falls far less than the alarm implies, and it recovers.

Two principles organise the paper.

Two principles

Deeds over words. When **Say** and **Do** disagree, trust **Do**. We show that the *sign* of the covariance between words and deeds decides whether words should be followed or faded.

Fade the echo, follow the news. Repetition adds attention but no information, so its price effect must unwind. New information that the crowd under-weights keeps moving the price.

Neither principle amounts to blind contrarianism. Prices drift after genuine news (Chan, 2003), so a rule that always fades the headlines loses money. The theory instead identifies which of the four cells in Figure 1b a given event belongs to.

What is new. The results we believe to be new are:

1. **An endogenous theory of false alarms** (Section 4). A closed-form model of optimal speech while trading shows exactly when an institution talks an asset down while buying it, and when it exaggerates while selling.
2. **The Say–Do covariance identity** (Theorem 4.4). It holds for any distribution and turns an observable statistic into a follow-or-fade switch.
3. **How virality shapes manipulation** (Proposition 4.5) and **how trust can cycle** (Theorem 4.6), with explicit thresholds.

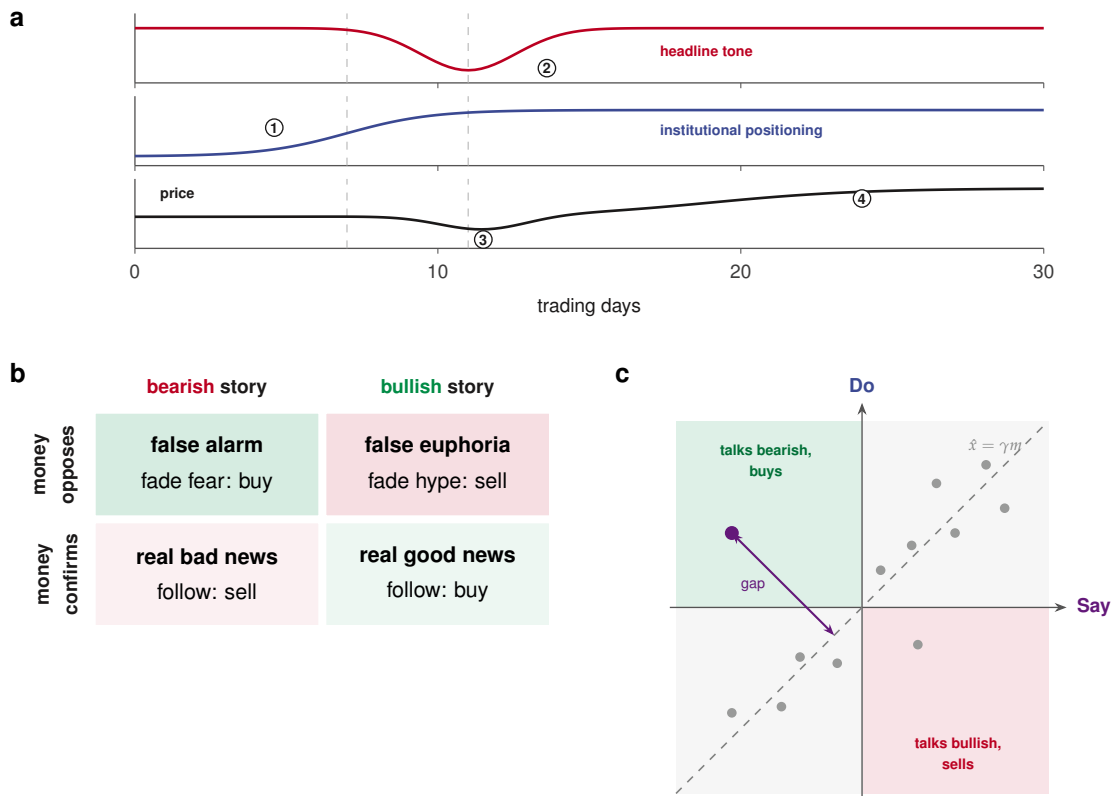


Figure 1 | Three voices, one decision. **a**, The false-alarm pattern (stylised): positioning builds (1) before the alarm (2); the price falls far less than the alarm implies (3), then recovers (4). **b**, The narrative alone never decides the trade; its agreement with money and price does. **c**, The Say–Do plane. The optimal forecast falls with the signed distance from the line $\hat{x} = \gamma m$ (Theorem 3.6); the top-left quadrant is the false-alarm zone.

4. **Semantic declustering** (Theorem 5.3): an exact way to separate news from echo that combines timing with meaning.
5. **Guarantees for market-aligned text embeddings** (Theorems 5.4 and 5.5), **absorption** as the gap between implied and actual price moves (Theorem 3.7), and **lead-lag** measured by Lévy area (Theorem 5.6).

The Gaussian results of Section 3 are clean multi-channel statements of Bayesian updating, and the tools for regime detection, calibration and position sizing (Appendix B) are classical. They are included so that the paper is self-contained.

How to read this paper. Section 3 builds the simplest market in which the three voices interact. Section 4 lets institutions choose what to say, which is where false alarms come from. Section 5 turns the theory into measurements, and Section 6 combines them into a decision. Section 7 checks every result numerically. Each main result is followed by a short plain-language reading, and all proofs are in Appendix A.

2 Background

Sentiment and prices. Media pessimism predicts short-term price pressure followed by reversal (Tetlock, 2007). Financial text needs domain-specific measurement (Loughran and McDonald, 2011). Sentiment matters most where arbitrage is hard and risky (De Long et al.,

1990; Shleifer and Vishny, 1997; Baker and Wurgler, 2006), and attention drives the demand of less-informed investors (Barber and Odean, 2008; Da et al., 2011). Under- and overreaction can coexist in one market (Hong and Stein, 1999), and the same news is read differently in different states of the world (Veronesi, 1999). Supervised text factors and language models now predict returns (Ke et al., 2019; Araci, 2019; Lopez-Lira and Tang, 2023), although pretrained models can leak future information into backtests (Glasserman and Lin, 2023).

Informed trading and strategic communication. Order flow reveals private information (Kyle, 1985; Glosten and Milgrom, 1985). Volume–return dynamics separate informed trades from hedging (Llorente et al., 2002). Sophisticated traders exploit predictable flow (Brunnermeier and Pedersen, 2005), short sellers process public news better than others (Engelberg et al., 2012), and promotional articles move prices (Kogan et al., 2023). On the communication side, cheap talk (Crawford and Sobel, 1982), costly talk with credulous listeners (Kartik et al., 2007), manipulation by informed “gurus” (Bénabou and Laroque, 1992) and trading on rumours (van Bommel, 2003) study how informed agents shape beliefs.

Tools. We use Hawkes processes (Hawkes, 1971) and their cluster representation (Hawkes and Oakes, 1974), which underlies stochastic declustering (Zhuang et al., 2002) and is standard in finance (Bacry et al., 2015). We also use contrastive representation learning (van den Oord et al., 2018; Reimers and Gurevych, 2019), the linear representation hypothesis (Park et al., 2024), Lévy area and path signatures (Lévy, 1940; Lyons, 1998; Chevyrev and Kormilitzin, 2016), cobweb learning and period-doubling (Ezekiel, 1938; Evans and Honkapohja, 2001; May, 1976), and the square-root impact law (Bouchaud et al., 2018).

3 A Market with Three Voices

We begin with the simplest market in which the three voices interact. The goal is not realism but *signs*: which signals should be followed, which should be faded, and why.

3.1 Setup

A stock has a true value v . The public receives several channels about it: an institutional statement, independent news, and media repetition. A crowd moves the price in response to these channels, while an informed trader who knows v trades against any mispricing. Figure 2 shows who influences whom.

Assumption 3.1 (Signals). Value is $v \sim \mathcal{N}(0, \sigma_v^2)$. The public channels are $\mathbf{y} = \mathbf{h}v + \boldsymbol{\nu} \in \mathbb{R}^K$, where $\boldsymbol{\nu} \sim \mathcal{N}(\mathbf{0}, \Sigma_\nu)$ is independent of v and $\Sigma_y = \sigma_v^2 \mathbf{h}\mathbf{h}^\top + \Sigma_\nu \succ 0$.

Assumption 3.2 (Prices). Noise traders submit $u \sim \mathcal{N}(0, \sigma_u^2)$, independent of everything else. Informed traders submit $x = \beta(v - p)$ with $\beta > 0$. The price is

$$p = \boldsymbol{\varphi}^\top \mathbf{y} + \lambda(x + u), \quad \lambda > 0,$$

where the vector $\boldsymbol{\varphi}$ collects the crowd’s *narrative multipliers*: how strongly the price responds to each channel.

Three quantities appear throughout. The *Bayes weights* $\mathbf{w}^* = \sigma_v^2 \Sigma_y^{-1} \mathbf{h}$ describe how a

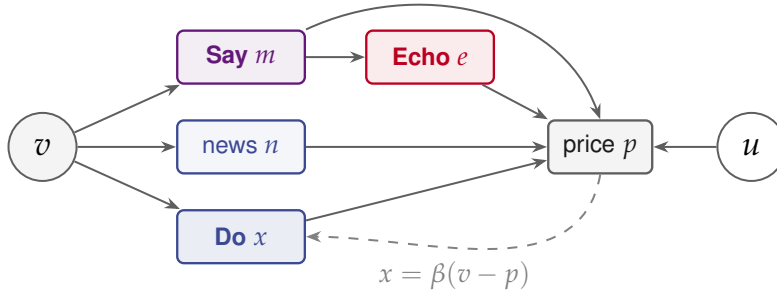


Figure 2 | Information structure. Echo is a downstream copy (a “garbling”) of institutional speech, news is an independent signal of value, and informed demand responds to the price (dashed).

perfectly rational observer would read the channels, so that $\mathbb{E}[v | \mathbf{y}] = \mathbf{w}^* \mathbf{y}$. The number $\kappa = (1 + \lambda\beta)^{-1} \in (0, 1)$ measures how much mispricing survives informed trading. And $\sigma_{v|y}^2 = \text{Var}(v | \mathbf{y})$ is the uncertainty that remains after reading the news. The forward return is $r = v - p$, which works out to $r = \kappa(v - \boldsymbol{\varphi}^\top \mathbf{y} - \lambda u)$.

3.2 Follow or fade

The first question is simple: after a piece of news, should we expect the price to keep going or to come back?

Theorem 3.3 (Follow or fade). *Under Assumptions 3.1 and 3.2,*

$$\mathbb{E}[r | \mathbf{y}] = \kappa (\mathbf{w}^* - \boldsymbol{\varphi})^\top \mathbf{y}, \quad \text{Var}(r | \mathbf{y}) = \kappa^2 (\sigma_{v|y}^2 + \lambda^2 \sigma_u^2) =: \varsigma^2.$$

Returns load negatively on channel k (fade) if and only if $\varphi_k > w_k^*$.

In words. Compare how much the crowd reacts to a channel with how much that channel is actually worth. If the crowd overreacts, the move reverses; if it underreacts, the move continues.

Two special cases carry the paper’s second principle.

Corollary 3.4 (Fade the echo). *If the echo channel is a garbling of the others, $e = \mathbf{a}^\top \mathbf{y}_{-e} + \xi$ with independent noise ξ of positive variance, then $w_e^* = 0$ and the return loading on e is exactly $-\kappa \varphi_e$.*

Corollary 3.5 (Follow the news). *If a news channel is an independent signal of value, $n = v + v_n$, then $w_n^* \in (0, 1)$, and returns drift in the direction of n whenever $\varphi_n < w_n^*$.*

In words. A repeated story is worth nothing beyond the original, so whatever it moves is borrowed and must be returned. A genuinely new story that the crowd under-prices keeps pushing the price in its direction.

3.3 Deeds over words

Suppose we can also observe, with noise, what informed capital is doing: a positioning proxy $\hat{x} = x + \xi_x$ with $\text{Var} \xi_x = \sigma_x^2$.

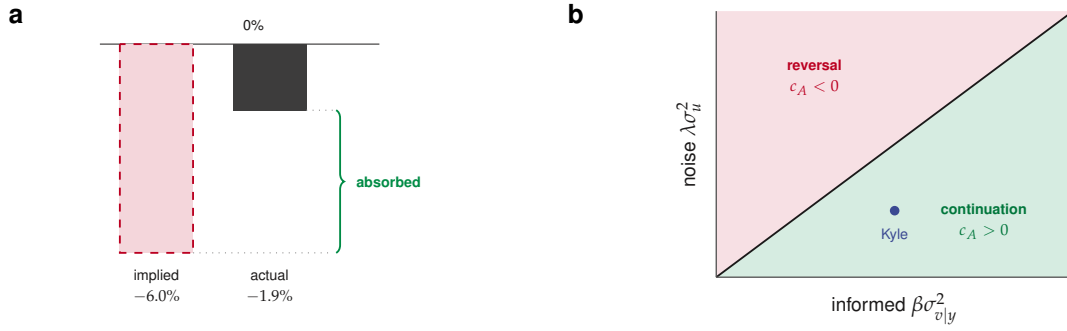


Figure 3 | Absorption. **a**, The narrative implied a 6% fall; the price fell 1.9%. **b**, Phase diagram of [Theorem 3.7](#): absorption signals a hidden buyer only where informed trading dominates noise. The dot marks Kyle's equilibrium.

Theorem 3.6 (Say–Do gap). Let $c_x = \beta\zeta^2 / (\beta^2\zeta^2 + \sigma_x^2)$ and $\theta = \sigma_x^2 / (\beta^2\zeta^2 + \sigma_x^2)$. Then

$$\mathbb{E}[r \mid \mathbf{y}, \hat{x}] = \theta\kappa(\mathbf{w}^* - \boldsymbol{\varphi})^\top \mathbf{y} + c_x \hat{x}, \quad \text{Var}(r \mid \mathbf{y}, \hat{x}) = \theta\zeta^2 < \zeta^2.$$

If the institutional channel $m = y_1$ is over-weighted by the crowd ($\varphi_1 > w_1^*$), the forecast depends on words and deeds only through $-c_x(\gamma m - \hat{x})$, with $\gamma = \theta\kappa(\varphi_1 - w_1^*)/c_x > 0$.

In words. Watching what money does always sharpens the forecast. When words are over-believed, the best forecast is a weighted *gap* between what is said and what is done, with an explicit exchange rate γ . “Talks bearish, buys” is bullish.

3.4 Absorption: did someone quietly buy?

The *absorption* of a narrative is the price move net of the move the narrative implied: $a = p - \boldsymbol{\varphi}^\top \mathbf{y}$. If the story implied a 6% fall and the price fell 2%, the missing 4% is absorption ([Figure 3a](#)).

Theorem 3.7 (Absorption sign). $\mathbb{E}[r \mid \mathbf{y}, a] = \mathbb{E}[r \mid \mathbf{y}] + c_A(a - \mathbb{E}[a \mid \mathbf{y}])$, where

$$c_A = \frac{\beta\sigma_{v|y}^2 - \lambda\sigma_u^2}{\lambda(\beta^2\sigma_{v|y}^2 + \sigma_u^2)}.$$

Absorption predicts continuation if and only if $\beta\sigma_{v|y}^2 > \lambda\sigma_u^2$, and reversal otherwise.

Lemma 3.8 (Robustness). For any estimate $\hat{\boldsymbol{\varphi}}$ of the multipliers, $\hat{a} = p - \hat{\boldsymbol{\varphi}}^\top \mathbf{y}$ carries exactly the same information as a once $\mathbf{y}$ is known.

In words. If informed buyers are the ones absorbing the selling, the price will follow them. If the absorption is just noise, it will fade. Which case holds depends on one comparison, shown in [Figure 3b](#). At the classical equilibrium of [Kyle \(1985\)](#), $c_A = \frac{1}{2}$: absorption is informative. The lemma means that we do not need to know the crowd's reaction exactly for the signal to work.

This result is close in spirit to [Llorente et al. \(2002\)](#), where high-volume returns continue when informed trading dominates and reverse when hedging dominates. The difference is that we condition on the *narrative*.

4 Why False Alarms Happen

So far the institution's statement was taken as given. We now let the institution *choose* what to say while it trades. This is where false alarms come from.

4.1 An institution that speaks while it trades

Assumption 4.1 (Strategic institution). An institution knows v , which may have any distribution with mean 0 and variance $\sigma_v^2 < \infty$. It publishes a message m and trades x . The public sees $\tilde{m} = m + \varepsilon$, where ε has mean zero and variance σ_ε^2 and is independent of v . The price is $p = \varphi \tilde{m} + \lambda x$, where $\varphi \geq 0$ is the crowd's *credulity*. Misreporting costs $\frac{k}{2}(m - v)^2$ (legal and reputational risk). The institution maximises

$$J(x, m) = x(v - \varphi m - \lambda x) - \frac{k}{2}(m - v)^2.$$

The institution faces a simple trade-off. Its message moves the price at which it trades, but lying is costly. One number summarises the market environment: the *deterrence–impact product* $a := 2\lambda k$.

Theorem 4.2 (Optimal speech and trade). *J has a unique maximiser if and only if $a > \varphi^2$. It is linear in value, $m^* = \psi v$ and $x^* = \chi v$, with*

$$\psi = \frac{a - \varphi}{a - \varphi^2}, \quad \chi = \frac{k(1 - \varphi)}{a - \varphi^2}.$$

If $a < \varphi^2$, the objective is unbounded: only position limits contain manipulation.

The slopes ψ and χ tell us how the institution speaks and trades per unit of good news. Their signs produce a complete taxonomy (Table 2 and Figure 4a).

Corollary 4.3 (Taxonomy of speech). *Let $a > \varphi^2$. The institution is in exactly one of the four regimes of Table 2.*

Table 2 | Four ways to speak while trading (for good news, $v > 0$).

Regime	Condition	Speech	Trade	What it looks like
truth	$\varphi = 1$	$\psi = 1$	$\chi = 0$	says what it knows, does not trade
shading	$\varphi < 1, a > \varphi$	$0 < \psi < 1$	buys	understates good news and buys
false alarm	$\varphi^2 < a < \varphi < 1$	$\psi < 0$	buys	talks the asset down while buying it
exaggeration	$\varphi > 1$	$\psi > 1$	sells	overstates and sells into the hype

In words. False alarms appear when the crowd is fairly credulous and lying is cheap relative to price impact, but not so cheap that manipulation becomes unbounded. They occupy a thin wedge of the parameter space. The mirror image, pumping and dumping, appears when the crowd over-believes ($\varphi > 1$).

4.2 Words, deeds and returns

The next result is the most useful one in practice, because it needs no model of how value is distributed.

Theorem 4.4 (Say–Do covariance identity). *Under [Assumption 4.1](#), let the message rule $m(v)$ be arbitrary, and let the institution trade optimally given its message. Then, path by path, $r = \lambda x - \varphi \varepsilon$. Hence, for any distribution of v ,*

$$\text{Cov}(r, \tilde{m}) = \lambda \text{Cov}(x, \tilde{m}) - \varphi \sigma_\varepsilon^2, \quad \mathbb{E}[r \mid x] = \lambda x.$$

In equilibrium, $\text{Cov}(x, \tilde{m}) = \psi \chi \sigma_v^2$, which is negative exactly in the false-alarm and exaggeration regimes.

Why the identity matters

Both sides can be estimated from data, and no distributional assumption is needed. The rolling sign of $\widehat{\text{Cov}}(\text{Say}, \text{Do})$ tells us whether institutional words should be followed or faded. Deeds are always followed, with slope equal to price impact.

4.3 How virality shapes manipulation

Media echo amplifies every message. If each article moves the price by φ_0 per unit of message and a Hawkes cascade with branching ratio $n \in [0, 1)$ repeats it, the expected cascade size is $(1 - n)^{-1}$ ([Proposition 5.1](#)). The effective credulity is therefore $\varphi = \varphi_0 / (1 - n)$. More virality means more credulity.

Proposition 4.5 (Manipulation dichotomy). *Let $\varphi_0 \in (0, 1)$ and call a regime manipulative if $\text{Cov}(\text{Say}, \text{Do}) < 0$.*

- (i) *In a deep market ($a < 1$), the only manipulative regime is the false alarm. It occurs if and only if $\max\{0, 1 - \varphi_0/a\} < n < 1 - \varphi_0/\sqrt{a}$.*
- (ii) *In a shallow market ($a > 1$), it is exaggeration. It occurs if and only if $1 - \varphi_0 < n < 1 - \varphi_0/\sqrt{a}$.*
- (iii) *Above $n = 1 - \varphi_0/\sqrt{a}$ no bounded optimum exists.*

In words. Virality does not create one kind of manipulation; it creates the kind the market structure allows. In liquid markets with low impact, echo makes talking down and buying optimal. In thin markets, it makes pumping and dumping optimal. Both happen inside explicit windows of the echo share ([Figure 4b](#)).

4.4 Trust that learns, and cycles

Suppose that after each round the crowd resets its credulity to what the last round's data suggest, namely the least-squares slope of value on message. This is a cobweb-type learning rule ([Ezekiel, 1938](#); [Evans and Honkapohja, 2001](#)), and it gives

$$\varphi_{t+1} = \frac{1}{\psi(\varphi_t)} = F_a(\varphi_t), \quad F_a(\varphi) = \frac{a - \varphi^2}{a - \varphi}.$$

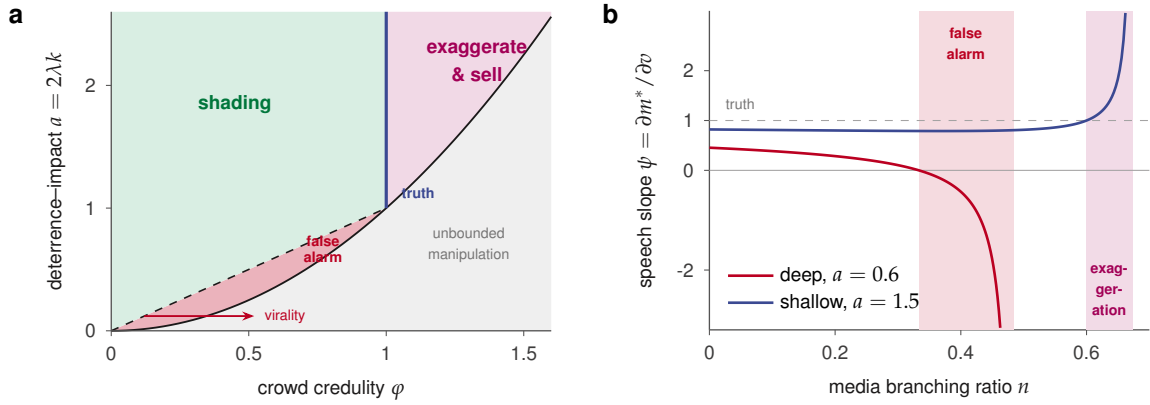


Figure 4 | Where false alarms live. **a**, Regime map (Theorem 4.2, Table 2). False alarms occupy the wedge $\varphi^2 < a < \varphi$; echo amplification moves a market to the right. **b**, Speech slope against the media echo share ($\varphi_0 = 0.4$, Proposition 4.5). A deep market passes through a false-alarm window where the slope turns negative; a shallow market passes through an exaggeration window where it exceeds one.

- Theorem 4.6** (Credibility cycles). (i) For $a > 1$, truth ($\varphi = 1$) is the only admissible resting point. (ii) It is stable if and only if $a > 2$, that is, $\lambda k > 1$. (iii) For $1 < a < 2$ there is exactly one 2-cycle, $\varphi_{\pm} = \frac{1}{2}(a \pm \sqrt{a(2-a)})$ with $\varphi_- < 1 < \varphi_+$. Its multiplier is $(5a - 9)/(a - 1)$, so it is stable if and only if $\frac{5}{6} < \lambda k < 1$. Along the cycle the market alternates between under-trust (shading) and over-trust (exaggeration). (iv) The cycle is born at $a = 2$ and loses stability at $a = \frac{5}{3}$, both through flip bifurcations.

In words. Trust settles on the truth only when deterrence is strong enough. With moderate deterrence, the crowd alternates between believing too little and believing too much, and institutional behaviour swings between shading and exaggeration. With weak deterrence, the swings become chaotic and eventually trust collapses (Figure 5). The testable prediction is that the rolling Say–Do covariance should change sign from period to period in such markets.

Bénabou and Laroque (1992) derive credibility dynamics from Bayesian learning about a random honest type, and Kartik et al. (2007) study costly talk with credulous listeners. The present model differs by coupling speech to trading in a linear–quadratic market, which makes the false-alarm region, the covariance identity and the bifurcation thresholds explicit and estimable.

5 Measuring the Three Voices

The theory needs three measurements: how much of the coverage is echo, how surprising a story is, and who moved first. This section provides each one with a guarantee.

5.1 Echo: separating news from repetition

Let articles about an asset arrive at times $\tau_1 < \dots < \tau_N$ with text embeddings z_j . We model arrivals as a Hawkes process, in which every article briefly raises the chance of further

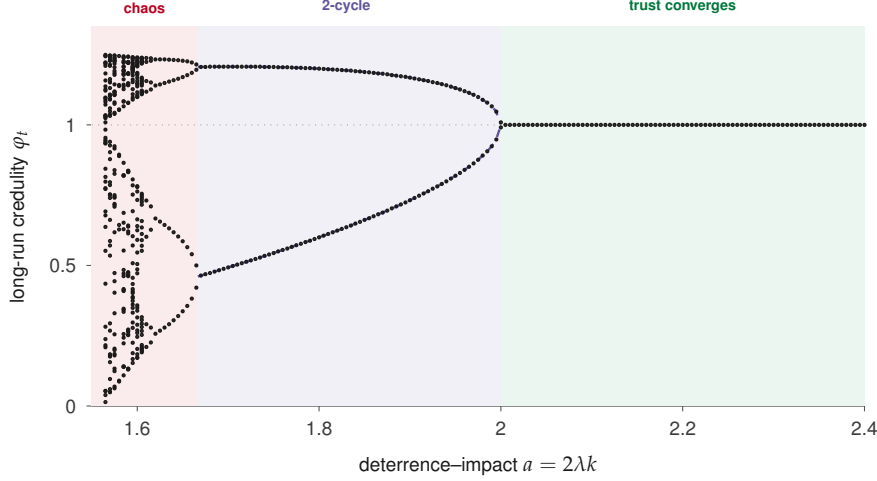


Figure 5 | How trust learns. Long-run credulity under adaptive trust (Theorem 4.6). Dots show the attractor; dashed curves show the closed-form cycle $\varphi_{\pm}$. Trust converges when $a > 2$, alternates when $\frac{5}{3} < a < 2$, and becomes chaotic below (numerical Lyapunov exponent ≈ 0.10 at $a = 1.6$). For $a \lesssim 1.56$ the orbit leaves the admissible set and credibility collapses.

articles:

$$\lambda(t) = \mu + \sum_{\tau_l < t} g(t - \tau_l), \quad n = \int_0^\infty g < 1.$$

In the cluster representation (Hawkes and Oakes, 1974), every article is either *news* (an “immigrant” arriving at rate μ) or an *echo* of an earlier article.

Proposition 5.1 (Echo share). *In the long run, the share of articles that are echoes equals the branching ratio n , and the average cascade has $(1 - n)^{-1}$ articles.*

Timing alone is a weak clue, because a reprint usually *looks like* its parent. We therefore let meaning enter as well.

Assumption 5.2 (Marked cascade). News embeddings are drawn independently from a density f_0 . An echo of article l has an embedding drawn from $f(\cdot | z_l)$, for example the von Mises–Fisher density $f \propto e^{\kappa \langle z, z_l \rangle}$, which concentrates near the parent.

Theorem 5.3 (Semantic declustering). *Given all arrival times and embeddings, the parent of each article is independent of the parents of the others, with*

$$\mathbb{P}(j \text{ is news} | \cdot) = \frac{\mu f_0(z_j)}{\Lambda_j}, \quad \mathbb{P}(j \text{ echoes } l | \cdot) = \frac{g(\tau_j - \tau_l) f(z_j | z_l)}{\Lambda_j},$$

where Λ_j is the sum of the numerators over all candidates. The best estimates of the news and echo sentiment totals are

$$s^{\text{new}} = \sum_j p_j s_j, \quad s^{\text{echo}} = \sum_j (1 - p_j) s_j, \quad p_j = \mathbb{P}(j \text{ is news} | \cdot).$$

In words. Each article receives an exact probability of being news, based on when it arrived and how similar it is to what came before. The computation is causal (it uses only earlier articles) and costs one pass over the data. Ignoring meaning ($\kappa = 0$) recovers classical stochastic declustering (Zhuang et al., 2002).

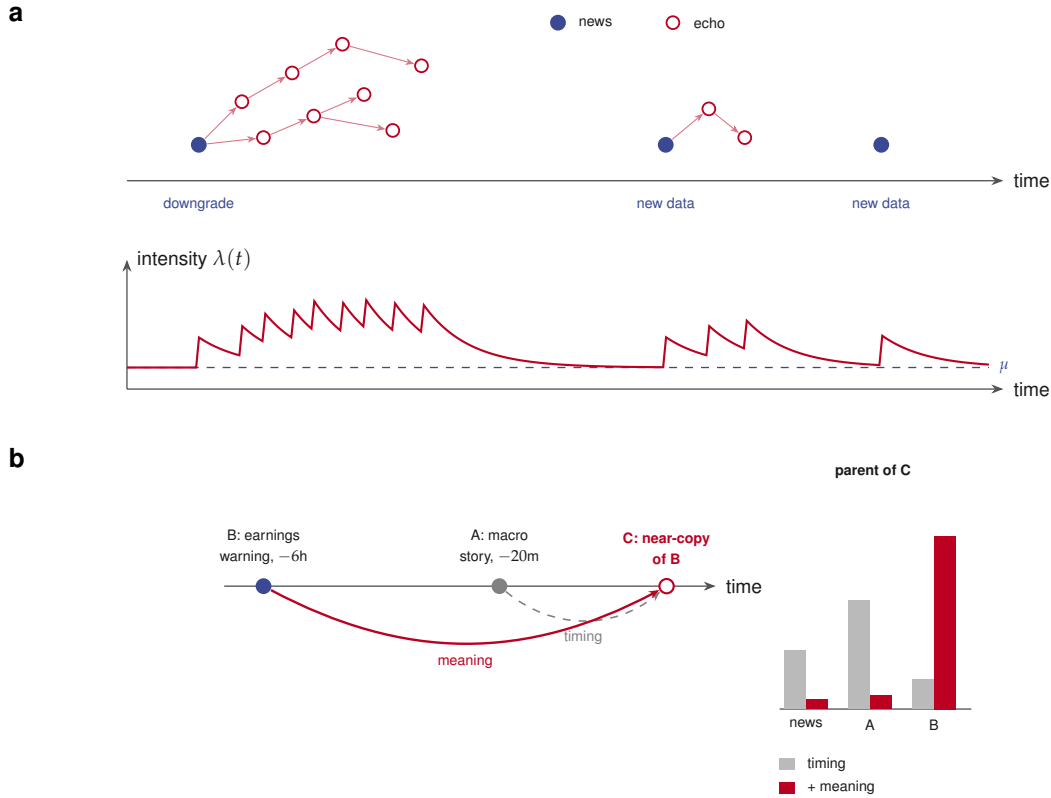


Figure 6 | Separating news from echo. **a**, A news article (blue) triggers a cascade of echoes (red); the arrival intensity jumps with each article and decays to the baseline μ . **b**, Semantic declustering (Theorem 5.3; illustrative numbers). Timing alone attributes article C to the most recent story, A. Adding meaning correctly identifies C as an echo of B.

5.2 A map of meaning aligned with markets

A text embedding places each document on a sphere, $z_j \in S^{D-1}$, so that similar texts lie close together. Generic embeddings cluster by *topic*. For our purposes they should cluster by *consequence*: stories that led to similar market reactions should be neighbours (Figure 7).

Let ρ_j collect what happened after story j (abnormal returns at several horizons, abnormal volume, and the change in implied volatility). Write $S_{jk} = \langle z_j, z_k \rangle$ and $D_{jk} = \|\rho_j - \rho_k\|^2$. The return-aligned contrastive loss asks the embedding to reproduce outcome similarities:

$$\mathcal{L}(Z) = - \sum_j \sum_{k \neq j} q_{jk} \log p_{jk}(Z), \quad q_{jk} \propto e^{-D_{jk}/2b^2}, \quad p_{jk}(Z) \propto e^{S_{jk}/\tau},$$

with both normalised over $k \neq j$.

Theorem 5.4 (Perfect alignment). $\mathcal{L}(Z) \geq \sum_j H(q_{j\cdot})$, with equality if and only if $S_{jk} = c - \frac{\tau}{2b^2} D_{jk}$ for a constant c . At equality, $\|z_j - z_k\|^2 = 2(1 - c) + \frac{\tau}{b^2} \|\rho_j - \rho_k\|^2$, so embedding neighbours are exactly outcome neighbours.

Training is never perfect, so the next result matters more in practice.

Theorem 5.5 (Certificate for imperfect training). For any embedding Z , let $\epsilon_j = \text{KL}(q_{j\cdot} \| p_{j\cdot}(Z))$ be the loss left over at story j . If $q_{jk} - q_{jl} > \sqrt{2\epsilon_j}$, then $\|z_j - z_k\| < \|z_j - z_l\|$.

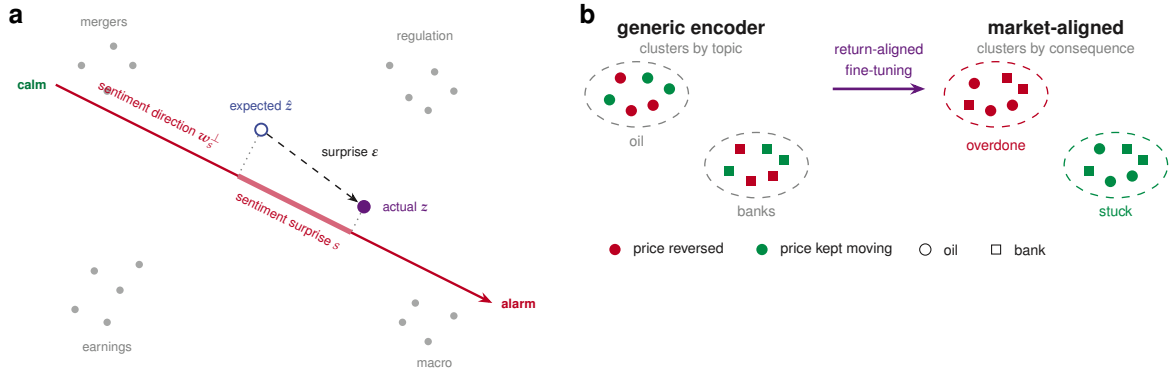


Figure 7 | A map of meaning aligned with markets. **a**, Stories cluster by topic, sentiment is a direction, and surprise is measured against the *expected* story; its projection onto the alarm direction is the sentiment surprise. **b**, At the optimum of the return-aligned loss (Theorem 5.4), distances mirror market aftermaths, so “news that got overdone” becomes a region of the map.

In words. The leftover training loss tells us which comparisons we can trust. Whenever two outcomes differ clearly enough, the embedding is guaranteed to rank them correctly. This is what makes “d  ja vu” forecasting safe: we estimate the reaction a new story implies by averaging what happened after its nearest historical neighbours (Proposition B.5), and the gap between that and the actual reaction is the absorption of Section 3.4.

Sentiment is read as a projection onto a direction $w_s^\perp = (I - UU^\top)w_s$ from which topic directions U have been removed, so that bad news in banking and bad news in energy are comparable. Surprise is measured against the *expected* next story $\hat{z}$, and its Mahalanobis size is χ^2 -calibrated (Proposition B.6).

5.3 Who moved first?

The strongest false alarms have a clear order: money moves first, and the story turns against it later. To measure order we use the *L  vy area* of two paths X and Y ,

$$\mathcal{A}_T = \frac{1}{2} \int_0^T (X_u - X_0) dY_u - (Y_u - Y_0) dX_u,$$

which is the signed area swept by the path (X, Y) and the antisymmetric part of its level-two signature.

Theorem 5.6 (Area identifies lead-lag). *Let (X, Y) be zero-mean, jointly stationary and mean-square differentiable, with cross-covariance $C(h) = \mathbb{E}[X_s Y_{s+h}]$. Then*

$$\mathbb{E}[\mathcal{A}_T] = T C'(0) - \frac{1}{2} (C(T) - C(-T)),$$

so the area grows at rate $C'(0)$. If $Y_t = \gamma X_{t-\ell} + Z_t$ with independent noise Z and a unimodal autocovariance for X , the rate has the sign of $\gamma\ell$.

In words. Plot positioning against headline gloom and follow the path over a month. If it loops one way, money led the story; if it loops the other way, money followed it. The enclosed area measures how strongly (Figure 8).

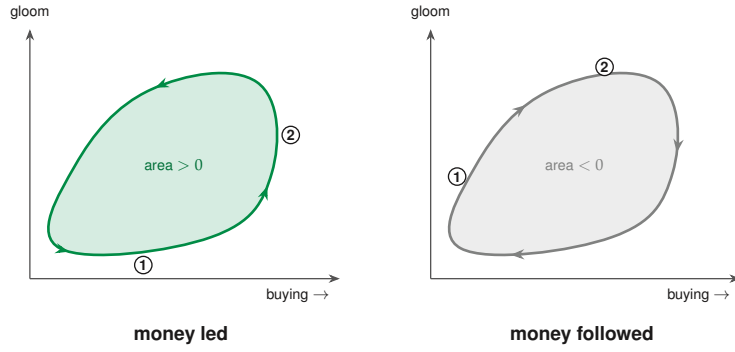


Figure 8 | Lead-lag as a loop. In the (positioning, gloom) plane, the direction of travel shows who moved first, and the enclosed area shows how strongly (Theorem 5.6).

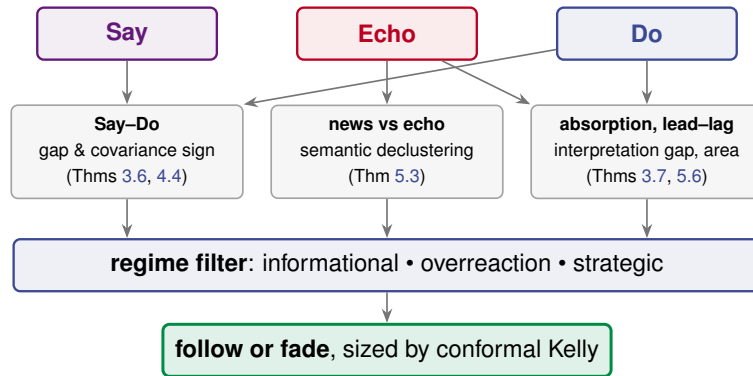


Figure 9 | The complete system. Three voices produce three families of theorem-signed signals; a regime filter turns them into one calibrated, sized decision.

6 From Signals to Decisions

Each event is assigned to one of three latent regimes: *informational* (money confirms the story), *overreaction* (mostly echo, no informed opposition) or *strategic* (money opposes the story and moved first). A hidden Markov filter updates the regime probabilities using only past data (Proposition B.1). The best forecast is then a probability-weighted mixture of regime-specific forecasts.

The individual signals combine into a transparent score in which every sign is fixed by a theorem:

$$D = \underbrace{\omega_1 s^{\text{new}}}_{\text{follow news}} - \underbrace{\omega_2 s^{\text{echo}}}_{\text{fade echo}} + \underbrace{\omega_3 \text{Do}}_{\text{follow deeds}} + \underbrace{\widehat{\text{sgn}} \omega_4 \gamma \text{Say}}_{\text{follow or fade words}} + \underbrace{\omega_5 A}_{\text{absorption}} + \underbrace{\omega_6 A^\pm}_{\text{money led}}.$$

Here $\widehat{\text{sgn}}$ is the rolling sign of $\text{Cov}(\text{Say}, \text{Do})$ (Theorem 4.4). When it is negative, the Say and Do terms combine into $-G_\gamma$, the Say-Do gap of Theorem 3.6. $A^\pm$ is the Lévy area signed by the direction of positioning.

Two safeguards turn the score into a position. Adaptive conformal intervals give forecast ranges whose long-run coverage holds for *any* sequence of outcomes (Theorem B.3). Positions are sized at a shrunk Kelly fraction $c^* = \text{SNR}/(1 + \text{SNR})$, because full Kelly loses money whenever the estimated edge is noisier than it is large (Theorem B.4). Figure 9 shows the complete system, and Table 3 walks through one event.

Table 3 | Illustration: a hypothetical downgrade. The numbers are illustrative, not data.

Signal (result)	Reading
news / echo sentiment (Theorem 5.3)	-0.5σ / -2.3σ
echo share n (Proposition 5.1)	0.82
Say–Do covariance (Theorem 4.4)	negative: fade the words
implied vs actual move, absorption (Theorem 3.7)	-6.0% vs -1.9%
Lévy area rate (Theorem 5.6)	positive: money led
regime probabilities, informational / overreaction / strategic	0.10 / 0.28 / 0.62
Kelly fraction (Theorem B.4) at SNR ≈ 0.5	$c^* \approx 0.33$
Reading: a probable false alarm; lean long at about one third of full Kelly.	

Table 4 | Theory versus independent checks (2×10^6 draws, seed 0).

Result	Quantity	Theory	Check
A market with three voices			
Corollary 3.4	Bayes weight on the echo channel	0	-0.0006
Theorem 3.3	return loading on the echo channel	-0.1840	-0.1843
Theorem 3.6	loading on positioning, c_x	0.4012	0.4004
Theorem 3.7	absorption coefficient, c_A	-0.1997	-0.2004
Why false alarms happen			
Theorem 4.2	speech slope ψ ($\varphi = 0.6, a = 0.5$)	-0.7143	-0.7143
Theorem 4.2	trade slope χ ($\varphi = 0.6, a = 0.5$)	1.4286	1.4286
Theorem 4.4	$\text{Cov}(r, \text{Say}), t_3$ values, false alarm	-2.294	-2.293
Theorem 4.4	$\text{Cov}(r, \text{Say}), t_3$ values, exaggeration	-7.608	-7.676
Theorem 4.6	cycle points at $a = 1.95$	$\{0.8189, 1.1311\}$	$\{0.8189, 1.1311\}$
Theorem 4.6	cycle multiplier at $a = 1.8$	0	9×10^{-16}
Measuring the voices			
Theorem 5.3	largest error of the factorised posterior	0	1.1×10^{-16}
Theorem 5.6	Lévy-area rate, lag 0.5	0.2356	0.2360
Decisions			
Theorem B.4	growth at the shrunk Kelly fraction	0.01056	0.01055
Theorem B.4	growth at full Kelly	-0.0300	-0.0300

7 Numerical Evidence

Every closed form in the paper was checked independently (Table 4). The Gaussian results were checked by regression on two million simulated draws. The Say–Do identity was checked with fat-tailed values (t_3) and Laplace noise, which tests its distribution-free claim. Optimal speech was compared with numerical optimisation, semantic declustering with brute-force enumeration of every possible parent assignment, and the Lévy-area rate with a smooth Gaussian process with a known lag. Theory values are exact, and all numbers are regenerated by the accompanying code.

The t_3 exaggeration case converges most slowly, because the fourth moment of a t_3 variable is infinite; the pathwise identity itself holds to machine precision.

Semantic declustering was also tested beyond exactness. We simulated news cascades in which echoes are von Mises–Fisher copies of their parents and asked each method to recover

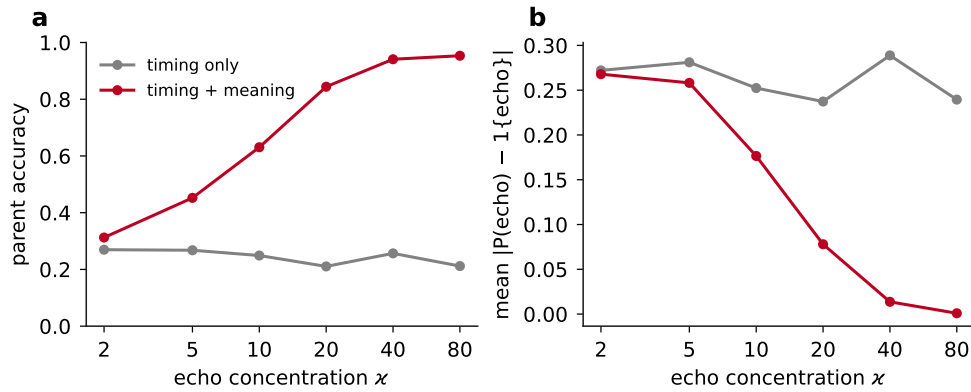


Figure 10 | Meaning makes echo detection work. Simulated cascades (Hawkes $\mu = 0.6$, $\alpha = 1$, $\beta = 1.25$; 16-dimensional embeddings). **a**, Accuracy of the most likely parent. **b**, Mean absolute error of each article's echo probability. Grey: timing only. Red: timing and meaning (Theorem 5.3).

the true parent of every article (Figure 10). Timing alone recovers about one parent in four, whatever the concentration. Adding meaning raises accuracy to about 95% when echoes are close copies, and it nearly eliminates the error in each article's probability of being an echo.

8 Discussion

Predictions. The theory makes sharp, testable claims:

- news sentiment and echo sentiment predict returns with *opposite* signs;
- the Say–Do covariance is negative in episodes that are later identified as manipulative;
- false alarms concentrate at intermediate echo shares in liquid markets, and exaggeration concentrates in illiquid ones;
- the rolling Say–Do covariance is negatively autocorrelated where deterrence is moderate;
- the absorption loading changes sign with the toxicity of order flow;
- returns are higher after episodes in which money moved first.

The cleanest first test pairs management tone on earnings calls (**Say**) with insider trades (**Do**). Both are public, and both concern the same actors.

Limitations. The crowd's pricing rule is reduced-form, and its credulity is fixed within a round. The Gaussian results pin down signs and sufficient statistics, not magnitudes. The lead–lag theorem assumes smooth, stationary paths, whereas real positioning and prices are rough. Above all, this paper contains no empirical backtest. Validation will require point-in-time data, purged cross-validation and deflated performance statistics (López de Prado, 2018; Bailey and López de Prado, 2014), as well as controls for look-ahead bias in pretrained language models (Glasserman and Lin, 2023).

Responsible use. A “strategic” reading means only that narrative and capital moved in opposite directions in a particular order. It is not evidence that anyone lied, and firms maintain information barriers between research and trading. The framework uses public information and lawful disclosures only, and it is intended for detection, research and

investor protection. Creating or spreading narratives intended to move prices is market manipulation. Nothing in this paper is investment advice.

Data and code availability. No proprietary data are used. A tested Python implementation of every result, together with scripts that regenerate [Table 4](#) and the quantitative figures, accompanies the paper at [repository URL].

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A Proofs

Lemma A.1 (Gaussian conditioning). *If $(\mathbf{a}, \mathbf{b})$ is jointly Gaussian with $\text{Var } \mathbf{b} \succ 0$, then $\mathbb{E}[\mathbf{a} \mid \mathbf{b}] = \mathbb{E}\mathbf{a} + \text{Cov}(\mathbf{a}, \mathbf{b}) \text{Var}(\mathbf{b})^{-1}(\mathbf{b} - \mathbb{E}\mathbf{b})$ and $\text{Var}(\mathbf{a} \mid \mathbf{b}) = \text{Var } \mathbf{a} - \text{Cov}(\mathbf{a}, \mathbf{b}) \text{Var}(\mathbf{b})^{-1} \text{Cov}(\mathbf{b}, \mathbf{a})$. The same holds conditionally on a further jointly Gaussian vector.*

Proof. The residual $\mathbf{e} = \mathbf{a} - \mathbb{E}\mathbf{a} - \text{Cov}(\mathbf{a}, \mathbf{b}) \text{Var}(\mathbf{b})^{-1}(\mathbf{b} - \mathbb{E}\mathbf{b})$ is uncorrelated with $\mathbf{b}$, and therefore independent of it by joint Gaussianity. Its variance expands to the stated formula. $\square$

Price and return. From $x = \beta(v - p)$, $p(1 + \lambda\beta) = \boldsymbol{\varphi}^\top \mathbf{y} + \lambda\beta v + \lambda u$, so $r = v - p = \kappa(v - \boldsymbol{\varphi}^\top \mathbf{y} - \lambda u)$.

Proof of Theorem 3.3. $\mathbb{E}[v \mid \mathbf{y}] = \sigma_v^2 \mathbf{h}^\top \Sigma_y^{-1} \mathbf{y} = \mathbf{w}^{\star\top} \mathbf{y}$ and $\mathbb{E}[u \mid \mathbf{y}] = 0$. The residual $r - \mathbb{E}[r \mid \mathbf{y}] = \kappa((v - \mathbb{E}[v \mid \mathbf{y}]) - \lambda u)$ has independent components given $\mathbf{y}$. $\square$

Proof of Corollary 3.4. $\sigma(\mathbf{y}) = \sigma(\mathbf{y}_{-e}, \xi)$ with ξ independent of $(v, \mathbf{y}_{-e})$, so $\mathbb{E}[v \mid \mathbf{y}] = \mathbb{E}[v \mid \mathbf{y}_{-e}]$, which is linear in $\mathbf{y}_{-e}$ alone. If $\mathbf{w}^{\star\top} \mathbf{y} = \mathbf{w}'^\top \mathbf{y}$ almost surely with $\mathbf{w}'_e = 0$, then $(\mathbf{w}^* - \mathbf{w}')^\top \Sigma_y (\mathbf{w}^* - \mathbf{w}') = 0$, and $\Sigma_y \succ 0$ gives $\mathbf{w}^* = \mathbf{w}'$. $\square$

Proof of Corollary 3.5. Conditionally on $\mathbf{y}_{-n}$, $v \sim \mathcal{N}(\mu', s^2)$ with $s^2 > 0$. The scalar update puts weight $s^2/(s^2 + \sigma_n^2) \in (0, 1)$ on n . Apply Theorem 3.3. $\square$

Proof of Theorem 3.6. $x = \beta r$, so $\hat{x} = \beta r + \xi_x$. Given $\mathbf{y}$, $\text{Cov}(r, \hat{x}) = \beta \zeta^2$ and $\text{Var } \hat{x} = \beta^2 \zeta^2 + \sigma_x^2$. Lemma A.1 gives $\mathbb{E}[r \mid \mathbf{y}, \hat{x}] = (1 - \beta c_x) \mathbb{E}[r \mid \mathbf{y}] + c_x \hat{x}$ with $1 - \beta c_x = \theta$, and $\text{Var} = \zeta^2 - c_x \beta \zeta^2 = \theta \zeta^2$. The $(m, \hat{x})$ terms are $-\theta \kappa(\varphi_1 - w_1^*) m + c_x \hat{x} = -c_x(\gamma m - \hat{x})$. $\square$

Proof of Theorem 3.7. Let $\tilde{v} = v - \mathbb{E}[v \mid \mathbf{y}]$. Then $r - \mathbb{E}[r \mid \mathbf{y}] = \kappa(\tilde{v} - \lambda u)$. Since $a = \lambda \beta r + \lambda u$, we get $a - \mathbb{E}[a \mid \mathbf{y}] = \lambda \beta \kappa \tilde{v} + \lambda(1 - \lambda \beta \kappa)u = \lambda \kappa(\beta \tilde{v} + u)$, using $1 - \lambda \beta \kappa = \kappa$. Hence $\text{Cov}(r, a \mid \mathbf{y}) = \lambda \kappa^2(\beta \sigma_{v|y}^2 - \lambda \sigma_u^2)$ and $\text{Var}(a \mid \mathbf{y}) = \lambda^2 \kappa^2(\beta^2 \sigma_{v|y}^2 + \sigma_u^2)$. Their ratio is c_A by Lemma A.1. At Kyle's intensities, $\beta \sigma_{v|y}^2 = \sigma_u \sigma_{v|y}$ and $\lambda \sigma_u^2 = \frac{1}{2} \sigma_u \sigma_{v|y}$, which give $c_A = \frac{1}{2}$. $\square$

Proof of Lemma 3.8. $\hat{a} = a + (\boldsymbol{\varphi} - \hat{\boldsymbol{\varphi}})^\top \mathbf{y}$ is an invertible affine map of a given $\mathbf{y}$. $\square$

Proof of Theorem 4.2. J is quadratic with Hessian $H = \begin{pmatrix} -2\lambda & -\varphi \\ -\varphi & -k \end{pmatrix}$, which is negative definite iff $\det H = a - \varphi^2 > 0$. Then J is strictly concave, and its unique maximiser solves $v - \varphi m - 2\lambda x = 0$ and $-\varphi x - k(m - v) = 0$. Substitution gives $m(a - \varphi^2) = v(a - \varphi)$, and $1 - \varphi \psi = a(1 - \varphi)/(a - \varphi^2)$ gives χ . If $\det H < 0$, then $J(t\mathbf{d}) \rightarrow \infty$ along an eigenvector $\mathbf{d}$ with positive eigenvalue. $\square$

Proof of Corollary 4.3. $\psi > 0 \iff a > \varphi$; $\psi < 1 \iff \varphi < 1$; and $\text{sgn } \chi = \text{sgn}(1 - \varphi)$. If $\varphi > 1$, then $a > \varphi^2 > \varphi$, so $\psi > 1$. $\square$

Proof of Theorem 4.4. Given m , the optimal trade is $x = (v - \varphi m)/(2\lambda)$, so $v - \varphi m = 2\lambda x$ and $r = v - \varphi(m + \varepsilon) - \lambda x = \lambda x - \varphi \varepsilon$. Since x and m are functions of v , which is independent of ε , we have $\text{Cov}(\varepsilon, \tilde{m}) = \sigma_\varepsilon^2$ and $\mathbb{E}[\varepsilon \mid x] = 0$. In equilibrium, $\text{Cov}(\chi v, \psi v) = \psi \chi \sigma_v^2$, and $\psi \chi < 0$ exactly in the false-alarm and exaggeration regimes. $\square$

Proof of Proposition 4.5. $\varphi = \varphi_0/(1-n)$ increases from φ_0 to ∞ . A false alarm needs $\varphi^2 < a < \varphi < 1$, hence $a < 1$. Exaggeration needs $1 < \varphi < \sqrt{a}$, hence $a > 1$. The conditions $\varphi > a$, $\varphi > 1$ and $\varphi < \sqrt{a}$ translate to $n > 1 - \varphi_0/a$, $n > 1 - \varphi_0$ and $n < 1 - \varphi_0/\sqrt{a}$. For $a < 1$, $\varphi < \sqrt{a}$ already implies $\varphi < 1$. $\square$

Proof of Theorem 4.6. (i) $F_a(\varphi) = \varphi \iff \varphi = 1$, and admissibility ($\varphi^2 < a$) needs $a > 1$. (ii) $F'_a(\varphi) = (a - 2a\varphi + \varphi^2)/(a - \varphi)^2$, so $F'_a(1) = -1/(a - 1)$, whose modulus is below 1 iff $a > 2$. (iii) Using $a - F_a(\varphi) = (a^2 - a\varphi - a + \varphi^2)/(a - \varphi)$, direct computation gives

$$F_a(F_a(\varphi)) - \varphi = \frac{-a(\varphi - 1)(2\varphi^2 - 2a\varphi + a^2 - a)}{(a - \varphi)^2(a - F_a(\varphi))}.$$

The quadratic factor has roots $\varphi_{\pm}$, which are real and different from 1 for $1 < a < 2$; its value at $\varphi = 1$ is $(a - 1)(a - 2) \neq 0$. They are admissible: $\varphi_{\pm}^2 = \frac{a}{2}(1 \pm \sqrt{a(2-a)}) < a$ because $a(2-a) < 1$; $\varphi_{\pm} < a$ because $\sqrt{a(2-a)} < a \iff a > 1$; and $\psi(\varphi_{\pm}) > 0$. On the cycle, $\varphi^2 = a\varphi - (a^2 - a)/2$, so the numerator of F'_a equals $a(3 - a - 2\varphi)/2$. With $\varphi_+ + \varphi_- = a$ and $\varphi_+\varphi_- = (a^2 - a)/2$, we get $(3 - a - 2\varphi_+)(3 - a - 2\varphi_-) = (5a - 9)(a - 1)$ and $(a - \varphi_+)(a - \varphi_-) = a(a - 1)/2$. Hence the multiplier is $(5a - 9)/(a - 1)$, whose modulus is below 1 iff $\frac{5}{3} < a < 2$ when $1 < a < 2$. (iv) At $a = 2$ the multiplier is 1 and $\varphi_{\pm} = 1$; at $a = \frac{5}{3}$ it is -1 . $\square$

Proof of Proposition 5.1. Each news article founds a Galton–Watson cluster of expected size $\sum_k n^k = (1 - n)^{-1}$. News arrives at rate μ , so articles arrive at rate $\mu/(1 - n)$, and the news share is $1 - n$. $\square$

Proof of Theorem 5.3. In the cluster representation, a labelled configuration is a realisation of independent Poisson streams: news at rate $\rho_0 = \mu$, and echoes of l at rate $\rho_l(t) = g(t - \tau_l)\mathbf{1}\{t > \tau_l\}$. Its density is the product of the stream likelihoods and the mark densities,

$$e^{-\int_0^T \lambda} \prod_j \rho_{\pi(j)}(\tau_j) f_{\pi(j)}(\mathbf{z}_j), \quad f_0 \text{ (news)}, \quad f_l = f(\cdot | \mathbf{z}_l),$$

because $\sum_k \rho_k = \lambda$. This factorises over j . Summing over labellings gives $e^{-\int \lambda} \prod_j \Lambda_j$, so the posterior is a product of the stated categorical laws. Because s_j is a function of $\mathbf{z}_j$, $\mathbb{E}[\mathbf{1}\{\pi(j) = 0\}s_j | \cdot] = s_j p_j^{\text{sem}}$. Integrating out the marks recovers the timing-only posterior, and conditioning does not increase expected entropy. $\square$

Proof of Theorem 5.4. Per anchor, $-\sum_k q_{jk} \log p_{jk} = H(q_{j\cdot}) + \text{KL}(q_{j\cdot} \| p_{j\cdot})$, with equality to $H(q_{j\cdot})$ iff $p_{j\cdot} = q_{j\cdot}$, that is, iff the logits differ by a constant c_j : $S_{jk}/\tau + D_{jk}/2b^2 = c_j/\tau$. Symmetry of S and D forces $c_j = c_k$ for every pair, and on the sphere $\|\mathbf{z}_j - \mathbf{z}_k\|^2 = 2 - 2S_{jk}$. $\square$

Proof of Theorem 5.5. Pinsker gives $\sum_k |p_{jk} - q_{jk}| \leq \sqrt{2\epsilon_j}$, so $p_{jk} - p_{jl} \geq q_{jk} - q_{jl} - \sqrt{2\epsilon_j} > 0$. Softmax is monotone in $S_{j\cdot}$, so $S_{jk} > S_{jl}$, which gives the distance ordering. $\square$

Proof of Theorem 5.6. Mean-square differentiability gives $\mathbb{E}[X_s \dot{Y}_t] = C'(t - s)$ and $\mathbb{E}[Y_s \dot{X}_t] = -C'(s - t)$. Hence $\mathbb{E}[(X_u - X_0)\dot{Y}_u] = C'(0) - C'(u)$ and $\mathbb{E}[(Y_u - Y_0)\dot{X}_u] = -C'(0) + C'(-u)$. By Fubini,

$$\mathbb{E}[\mathcal{A}_T] = \frac{1}{2} \int_0^T (2C'(0) - C'(u) - C'(-u)) du = TC'(0) - \frac{1}{2}(C(T) - C(-T)).$$

For $Y_t = \gamma X_{t-\ell} + Z_t$, $C(h) = \gamma R(h - \ell)$, so $C'(0) = -\gamma R'(\ell)$. This has the sign of $\gamma\ell$ when R is even with $R' < 0$ on $(0, \infty)$. $\square$

B Additional Results

Proposition B.1 (Causal filtering and mixture optimality). *Filtered regime probabilities $\pi_t \propto f(\mathbf{o}_t) \odot P^\top \pi_{t-1}$ are $\mathcal{F}_t$ -measurable, and $\mathbb{E}[r_{t+h} \mid \mathcal{F}_t] = \sum_k \pi_t(k) \mathbb{E}[r_{t+h} \mid \mathcal{F}_t, K_t = k]$. Smoothed probabilities are not $\mathcal{F}_t$ -measurable in general and must not be used for forecasting.*

Proof. The recursion is Bayes' rule, and the mixture follows from the tower property. $\square$

Theorem B.2 (Exponential detectability). *For n events that are i.i.d. within a regime, with emissions $\mathcal{N}(\mathbf{m}_i, \Sigma)$ and priors π_i , the MAP error satisfies $P_e \leq \sqrt{\pi_0 \pi_1} e^{-n\Delta^2/8} \leq \frac{1}{2} e^{-n\Delta^2/8}$, where $\Delta^2 = (\mathbf{m}_1 - \mathbf{m}_0)^\top \Sigma^{-1} (\mathbf{m}_1 - \mathbf{m}_0)$.*

Proof. $\min\{a, b\} \leq \sqrt{ab}$ gives $P_e \leq \sqrt{\pi_0 \pi_1} \text{BC}^n$ (Bhattacharyya, 1943; Kailath, 1967). For equal-covariance Gaussians, completing the square gives $\text{BC} = \int \sqrt{p_0 p_1} = e^{-\Delta^2/8}$. $\square$

Theorem B.3 (Adaptive conformal coverage; Gibbs and Candès, 2021, Prop. 4.1). *With nested sets $\hat{C}_t(\alpha')$ that equal $\mathbb{R}$ for $\alpha' \leq 0$ and $\emptyset$ for $\alpha' \geq 1$, and the update $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, every outcome sequence satisfies $|\frac{1}{T} \sum_t \text{err}_t - \alpha| \leq (\max\{\alpha_1, 1 - \alpha_1\} + \gamma)/(\gamma T)$.*

Proof (reproduced). By induction, $\alpha_t \in [-\gamma, 1 + \gamma]$. Telescoping gives $\alpha_{T+1} - \alpha_1 = \gamma \sum_t (\alpha - \text{err}_t)$. $\square$

Theorem B.4 (Kelly shrinkage under estimation risk; classical). *With excess growth rate $g(f) = f\mu - \frac{1}{2}f^2\sigma^2$ (Merton, 1969; Kelly, 1956), an unbiased estimate $\hat{\mu}$ with variance s^2 that is independent of future returns, and $f = c\hat{\mu}/\sigma^2$, the expected growth is $\mathbb{E}g = (c\mu^2 - \frac{1}{2}c^2(\mu^2 + s^2))/\sigma^2$. It is maximised at $c^* = \mu^2/(\mu^2 + s^2)$. Full Kelly yields $(\mu^2 - s^2)/(2\sigma^2)$, which is negative when $s > |\mu|$.*

Proof. Expand $\mathbb{E}[f\mu]$ and $\mathbb{E}[f^2]$ and maximise the resulting concave quadratic in c . $\square$

Proposition B.5 (Analog forecasting). *If $\text{AR}_j = \mu(z_j) + e_j$ with uncorrelated mean-zero errors of variance at most σ^2 , μ is L -Lipschitz, and weights $\omega_j \geq 0$ sum to one, then the estimator $\hat{\mu}(z) = \sum_j \omega_j \text{AR}_j$ has bias at most $L \sum_j \omega_j d(z, z_j)$ and variance at most $\sigma^2 \sum_j \omega_j^2$.*

Proposition B.6 (Calibrated surprise). *If $z \mid \mathcal{F} \sim \mathcal{N}(\hat{z}, \Sigma)$, then $S = (z - \hat{z})^\top \Sigma^{-1} (z - \hat{z}) \sim \chi_D^2$, and S equals Shannon surprisal up to additive constants.*

C Implementation Notes

Encoder. A transformer sentence encoder (Reimers and Gurevych, 2019) with low-rank adapters, trained on the return-aligned loss plus a KL anchor to the frozen encoder, on rolling windows whose outcome horizons close before each refit date. Pretrained components must predate each test window (Glasserman and Lin, 2023).

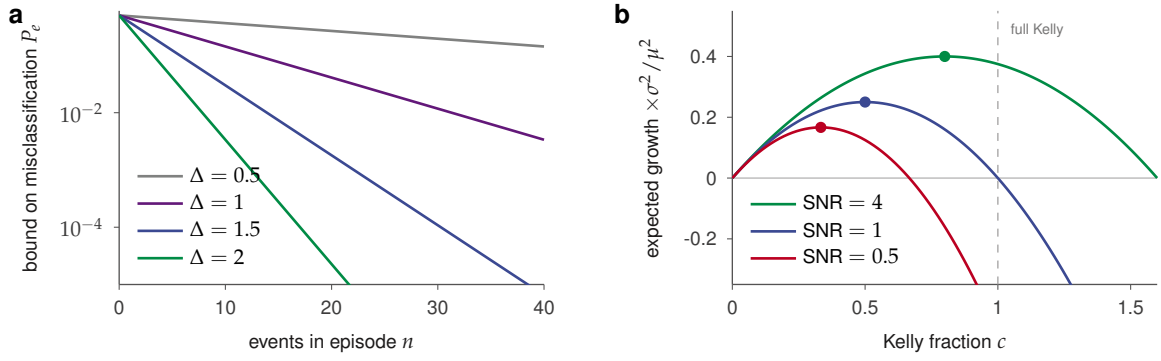


Figure 11 | Detection and sizing. **a**, Misclassification of strategic episodes decays exponentially in episode length (Theorem B.2). **b**, Expected growth as a function of the Kelly fraction; dots mark $c^* = \text{SNR} / (1 + \text{SNR})$ (Theorem B.4).

Echo. Exponential Hawkes kernels are fitted per semantic cluster, with von Mises–Fisher echo marks. The posteriors of Theorem 5.3 cost $O(N \cdot \bar{k})$ with $\bar{k}$ candidate parents retrieved from a vector index.

Say–Do. Pair management tone with insider trades, and published house views with disclosed holdings or futures positioning. A lag-augmented Kalman filter merges disclosures of mixed frequency, and a rolling window estimates $\text{Cov}(\text{Say}, \text{Do})$.

Validation. Point-in-time data, combinatorial purged cross-validation and deflated performance statistics (López de Prado, 2018; Bailey and López de Prado, 2014).